FACULTY OF ENGINEERING
CHULALONGKORN UNIVERSITY
2110201 Computer Engineering Mathematics
Year II, Second Semester, Midterm Examination, Mar 8, 2023 08:30-11:30

Nae.....Student ID..... CR58 .....

Remarks

1. **There are 24 questions in this exam paper, with the total number of 10 pages including this page.**
2. **The exam duration is 180 minutes.**
3. Write your Student ID, full name in the space provided on the top of every page.
4. No documents or calculators are allowed inside the exam room except for non-programmable and non-graphic calculators.
5. Borrowing is not allowed unless it is supervised by the proctor.
6. You must not bring any part of this exam paper outside. The exam paper is a government's property. Violators will be prosecuted under a criminal court.
7. Student who wishes to leave the exam room before the end of the exam period, must raise his/her hand and ask for permission before leaving the room. Student must leave the room in the orderly manner.
8. Once the time is expired, student must stop writing and must remain seated quietly until the proctors collect all the exam papers or given exam booklets. Only then, the students will be allowed to leave the room in the orderly manner.
9. Any student who does not obey the regulations listed above will receive punishment under the Faculty of Engineering Official Announcement on January 6, 2003 regarding the exam regulations.
   a) **With implicit evidence or showing intention for cheating, student will receive an F in that subject and will receive an academic suspension for 1 semester.**
   b) **With explicit evidence for cheating, student will receive an F in that subject and will receive an academic suspension for 1 year.**

ห้ามนิสิตพกโทรศัพท์และอุปกรณ์สื่อสารไว้กับตัวระหว่างสอบ หากตรวจพบจะถือว่านิสิตกระทำผิดเกี่ยวกับการสอบ อาจต้องพ้นสภาพการเป็นนิสิต หรือ ให้ได้รับ F และอาจพิจารณาให้ถอนรายวิชาอื่นทั้งหมดที่ลงทะเบียนไว้ในภาคการศึกษานี้

*** ร่วมรณรงค์การไม่กระทำผิดและไม่ทุจริตการสอบที่คณะวิศวกรรมศาสตร์ ***

ข้าพเจ้ายอมรับในข้อกำหนดที่กล่าวมานี้ ข้าพเจ้าเป็นผู้ทำข้อสอบนี้ด้วยตนเองโดยมิได้รับการช่วยเหลือ หรือให้ความช่วยเหลือ ในการทำข้อสอบนี้

1. Pencil is allowed.
2. Write your answer in the provided space only. If the space is not enough, you must use the back of the same page only.

ลงชื่อนิสิต.....

วันที่.....

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Part I: Basic Skill 1 & 2, provide detailed answers.

1. (2 points) Find the reduced row echelon of the following matrices.

   a. $\begin{bmatrix} 2 & 3 & 2 & 0 \\ 5 & 2 & 5 & 0 \\ 3 & 8 & 8 & 2 \end{bmatrix}$

   b. $\begin{bmatrix} 2 & 4 & 2 & 2 \\ 3 & 8 & 3 & 8 \\ 3 & 2 & 3 & 3 \end{bmatrix}$

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2. (2 points) Find the solution <u>set</u> of the systems of linear equations defined by the following augmented matrices. Compute the augmented matrices in RRE form. Identify homogeneous and particular parts of the solution. State which are pivot and free variables.

a. $\left[\begin{array}{cccccc} 1 & 3 & 2 & 0 & 4 & 1 \\ 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 3 \end{array}\right]$

b. $\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 2 & 4 & 8 \end{array}\right]$

c. $\left[\begin{array}{ccccccc} 0 & 2 & 0 & 3 & 0 & 0 & 5 \\ 0 & 0 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right]$

d. $\left[\begin{array}{ccc} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{array}\right]$

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3. (2 points) Find the determinant of the following matrices using row operations.

a. $\begin{bmatrix} 2 & -4 & 2 \\ 4 & -3 & 1 \\ 0 & -6 & 6 \end{bmatrix}$ b. $\begin{bmatrix} 0 & 2 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 4 & 1 \\ 1 & 0 & -1 & 0 & 5 \\ 0 & 0 & -3 & 1 & 1 \end{bmatrix}$

4. (1 points) Find a linear equation describing the plane containing the points (2,1,2,1), (4,3,2,1), (1,2,2,4) and (1,1,1,1) in $\mathbb{R}^4$.

5. (2 points) Find the inverse of the matrix using row operations.

$$\begin{bmatrix} -3 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 2 & -1 & 0 & 1 \end{bmatrix}$$

6. (1 point) Write the system of linear equations defined by the following augmented matrix.

$$\left[\begin{array}{cccccc} 3 & 2 & 2 & 0 & 1 & 1 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 0 & 1 & 0 & 2 & 1 & 3 \end{array}\right]$$

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Part II (1 point per question): Provide short answer only.

1. A mapping $F : \mathbf{R}^2 \to \mathbf{R}^3$, is given by the rule $F(x, y) = (x, y, y)$. Is F one-to-one, onto, expand or contract?

2. What is the angle between **v** and **w** when **v** = (1, 1, 0, 1, 1) and **w** = (1, 0, 0, 0, 1) in $\mathbf{R}^5$?

3. Find a linear equation whose solution set is the line through (2, 3) and perpendicular to (1, -2).

4. Suppose we have geometric vectors ***a*** and ***p*** in space with numeric coordinates **a** = (1, 1, 1) and **p** = (2, -1, -3), respectively. What equation describes the plane ***P*** orthogonal to the line generated by ***a*** and containing the tip of ***p***?

5. A mapping $F : \mathbf{R}^4 \to \mathbf{R}^2$ is given by the rule $F(x, y, z, w) = (x - y, z + w)$. Is F one-to-one and/or onto?

6. Let $A = \begin{bmatrix} 1 & 4 & 2 & 1 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 6 & 1 \\ 4 & 1 & 8 & 1 \end{bmatrix}$. Find a set of two linearly independent eigenvectors of A.

7. Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \\ 16 & 17 & 18 & 19 & 20 \end{bmatrix}$. Suppose that $A = BC$ where B is an $m \times n$ matrix and C is a $p \times q$ matrix. Find the range of possible values of m, n, p, q, and the relationship between these values.

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8. Let A be a 3×3 matrix and let $\boldsymbol{v} = \begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$. Given that $A\boldsymbol{v} = -\boldsymbol{v}$ and $A\boldsymbol{w} = 2\boldsymbol{w}$, find $A^5 \begin{bmatrix} 2 \\ -7 \\ -9 \end{bmatrix}$.

9. Let $S = \left\{ \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \middle| \text{there exist } x, y, z \in \mathbb{R} \text{ such that } \begin{matrix} x + 2y + 3z = b_1 \\ y - z = b_2 \\ x + 3y + 2z = b_3 \end{matrix} \right\}$. Does S form a subspace? If so, find a basis of S.

10. Let $A = \begin{bmatrix} 101 & 102 & 103 & 104 & 105 & 106 \\ 102 & 103 & 104 & 105 & 106 & 107 \\ 103 & 104 & 105 & 106 & 107 & 108 \\ 104 & 105 & 106 & 107 & 108 & 109 \\ 105 & 106 & 107 & 108 & 109 & 110 \\ 106 & 107 & 108 & 109 & 110 & 111 \end{bmatrix}$. Find $\det(A)$.

11. Let S_{ij} be an n x n identity matrix whose i[th] column is swapped with its jth column. Let A be an n x n matrix whose determinant is k. What is the value of $\min_{i,j \in \{1..n\}} (\det(S_{ij}A))$.

12. Let A be a matrix with n rows and n-1 columns. Let $y = Ax$ for some value of x and let B be the new matrix generated by appending y as the last column of A so that B is n x n matrix. How many different values of y makes $\det(B) = 0$

13. Write a matrix having more columns than rows and that $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is a basis of its column space.

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14. Give an example of a subspace in $\mathbf{R}^4$ such that there exists an explicit representation of that subspace where testing for a membership of that explicit representation is very easy. Write the answer as an implicit representation of that subspace.

15. There is an unknown matrix A and a matrix B where $B = \begin{bmatrix} -12 & 0 & 0 \\ -12 & -12 & 0 \\ -12 & -12 & -12 \end{bmatrix}$. Let the row space of BA be $\{(6,6,6)\}^{\perp}$, write the basis of the row space of A.

Part III: Provide detailed answer.

1. (5 points) Prove that a hyperplane in $\mathbb{R}^4$ that goes through the origin forms a subspace.

2. (5 points) Consider three planes in space that go through the origin and are respectively perpendicular to (1, 0, -1), (1, 1, 1), and (0, 1, -1). Determine whether these planes intersect in just the origin, an entire line, or a complete plane.

3. (5 points) In this problem you need to find a **<u>set</u>** of matrices $\boldsymbol{A}$ that satisfies all the given conditions. Justify your answer as well.
   a. $\boldsymbol{A}$ is 2 x 4 matrix.
   b. The left nullspace of $\boldsymbol{A}$ is the smallest subspace in $\mathbf{R}^2$
   c. The first column of $\boldsymbol{A}$ is a linear combination of the other columns of $\boldsymbol{A}$
   d. The vector (10,0,2,0) and (8,0,-4,0) are both in the nullspace of $\boldsymbol{A}$.
   e. The last row of $\boldsymbol{A}$ and $\mathrm{RRE}(\boldsymbol{A})$ is the same.
   f. Span of the last column of $\boldsymbol{A}$ is in the subspace defined by $\{(-2,-4)\}^{\perp}$